

# Mengxin Yang

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## Education

**Zhongnan University of Economics and Law**

Wuhan, Hubei

*B.S. in Statistics*

Sep 2023 – Jun 2027 (Expected)

- **GPA:** 3.58 / 4.00 (87.93 / 100)
- **Core Courses:** Natural Language Processing (98), Numerical Computing Methods (98), Unix and Linux Systems (98), Python Data Analysis (96), Mathematical Analysis (90), Multivariate Statistical Analysis (90), Machine Learning (88).

## Research Interests

AI for Science (AI4S), Biomolecular Foundation Models, Protein Conditional Generation, Protein Mutation Effect Prediction.

## Publications

- 1 Ziyu Shi, Yuxiang Tang, **Mengxin Yang**, Yunxin Xu\*. “MutCleaner: A Scalable Automated Cleaning Pipeline for Heterogeneous Protein Mutation Data.” *Under review at Bioinformatics*, 2026.
- 2 **Mengxin Yang**, Zimu Yu, Yunxin Xu\*. “Explicit Modeling of Protein Epistasis via L20R and ELEVar Architectures.” *Under review*, 2026.

*\*Note: The asterisk indicates the corresponding author.*

## Research Experience

### Protein Mutation Effect Prediction Incorporating Prior Knowledge

Mar 2026 – Sep 2026

*Co-Researcher* | Advisor: Assoc. Prof. Yunxin Xu

- Addressing the issue of existing models relying on black-box feature mappings, explored and designed the L20R and ELEVar network architectures with collaborators.
- **Outcome:** Attempted to introduce prior knowledge from statistics and structural biology to explicitly model protein epistasis from a mathematical perspective. This enhanced the interpretability of the model and achieved solid predictive performance on several public benchmarks.

### Protein Conditional Generative Framework Based on Flow Matching

Dec 2025 – Present

*Researcher* | Advisor: Assoc. Prof. Yunxin Xu

- Collected and processed million-scale protein sequence and structure data, utilizing MMseqs2 and Foldseek for sequence clustering and structural preprocessing.
- Participated in building a generative framework integrating Geometric Graph Neural Networks (GVP-GNN) and Flow Matching.
- **Outcome:** Preliminarily achieved protein sequence generation under given physical constraints (e.g., target  $T_m$ , functional constraints), successfully running the complete computational pipeline from data preprocessing to model generation.

### Automated Cleaning Tool for Protein Mutation Data (MutCleaner)

Mar 2026 – Aug 2026

*Co-Developer* | Advisor: Assoc. Prof. Yunxin Xu

- Reviewed and compared the formats and label definitions of multi-source public mutation datasets, and participated in the development of the automated data cleaning tool **MutCleaner** to address data heterogeneity.

- **Outcome:** Achieved validity verification and format standardization for multi-source data, constructing unified training and testing sets for deep learning models. This tool has been published on PyPI and is currently under review at *Bioinformatics*.

**Academic Service & Affiliations**

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**Xu Lab** | School of Statistics and Mathematics, Zhongnan University of Economics and Law  
Oct 2025 – Present

- Undergraduate Researcher, primarily responsible for maintaining the group’s data pipelines, scheduling server GPU resources, reproducing algorithms, and training models.

*Structures* (A Cell Press Journal) 2026

- Sub-reviewer: Assisted the advisor in peer review, primarily verifying the rationality of the data processing pipelines and statistical methods in the manuscripts.

**Skills**

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- **Programming & Environments:** Python, PyTorch, C++, R, Linux Shell, Git, LaTeX.
- **Bioinformatics Tools:** MMseqs2, Foldseek, PyMOL, DIAMOND, Seqkit, ESMFold.

**Honors & Awards**

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- **Successful Participant (S Prize),** Mathematical Contest in Modeling (MCM) 2026
- **Second Prize at Provincial Level,** 14th MathorCup Mathematical Modeling Challenge 2024
- **Third Prize at Provincial Level,** 16th Chinese Mathematics Competitions (CMC) 2024